

Student Number:

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2015
YEAR 11
HSC Assessment Point 1

Mathematics

General Instructions

- Date of task – Tuesday 17th November
- Weighting – 15%
- Reading time - 3 minutes
- Working time – 45 minutes
- Write using black or blue pen
- BOSTES approved calculators may be used
- A reference sheet is provided at the back of this paper
- Show all necessary working

Total marks - 34

Question 1-3

34 marks

- Start each question in a new booklet

Outcomes

- Tangent to a Curve and Derivative of a Function
- The Quadratic Polynomial
- Locus and the Parabola

Question 1 (10 Marks)

- a) Find the equation(s) of a point $P(x,y)$ that moves so that it is always 7 units from the line $y = 2$. 2
- b) Differentiate $5x^3 - 2x + 4$ with respect to x . 1
- c) Find the equation of the tangent to the curve $y = x^3 - 3x^2$ at the point $(3, 0)$. 3
- d) Find the equation of the directrix of the parabola $x^2 = -18y$. 2
- e) Find the values of k for which the quadratic equation $2x^2 + 8x + k = 0$ has 2 distinct, real roots. 2

Question 2 (14 Marks) Start this question in a new booklet

- a) Differentiate the following with respect to x , leaving your answer in fully factored form:
- (i) $\frac{x-2}{x+3}$ 2
- (ii) $x^3(2x-9)^5$ 3
- b) Let α and β be the solutions of $x^2 + 12x + 3 = 0$. Find:
- (i) $\alpha + \beta$ 1
- (ii) $\alpha\beta$ 1
- (iii) $\frac{1}{\alpha} + \frac{1}{\beta}$ 2
- (iv) $\alpha + \frac{3}{\alpha}$ 2
- c) Find the values of a , b and c , given that $3x^2 - 14x - 2 \equiv a(x-2)^2 + b(x+4) + 1 + c$ 3

Question 3 (10 Marks) Start this question in a new booklet

- a) Given that $f(x) = \sqrt{4x+b}$ and that $f'(11) = \frac{2}{5}$,
- (i) Show that $f'(x) = \frac{2}{\sqrt{4x+b}}$ 2
- (ii) Hence, find the value of b . 2
- b) Consider the parabola $y^2 - 4y - 6x - 8 = 0$.
- (i) Find the coordinates of the vertex 2
- (ii) Find the coordinates of the focus 2
- (iii) Show that the line $3x + 4y - 3 = 0$ intersects the parabola exactly twice. 2

End of Examination

Question 1		
Q	Solution	Suggested marking criteria
A	$y = 9$ and $y = -5$	1 mark for $y = 9$ 1 mark for $y = -5$
B	$\frac{d}{dx}(5x^3 - 2x + 4) = 15x^2 - 2$	1 mark for correct answer
C	$y = x^3 - 3x^2$ $y' = 3x^2 - 6x$ When $x = 3$, $y' = 9$ $y - 0 = 9(x - 3)$ $y = 9x - 27$	1 mark for $y' = 3x^2 - 6x$ 1 mark for $y' = 9$ at $x = 3$ 1 mark for $y = 9x - 27$ or equivalent
D	$4a = 18$ $a = \frac{18}{4}$ $= \frac{9}{2}$ Directrix equation is $y = \frac{9}{2}$	1 mark for the focal length, a 1 mark for the equation of the directrix
E	Two distinct real roots $\therefore \Delta > 0$ $\Delta = 8^2 - 4 \times 2 \times k$ $= 64 - 8k$ $\therefore 64 - 8k > 0$ $-8k > -64$ $k < 8$	1 mark for $64 - 8k > 0$ 1 mark if answer follows

SECTION 2		
Q	Solution	Suggested marking criteria
A i	$\frac{d}{dx}\left(\frac{x-2}{x+3}\right) = \frac{1(x+3) - 1(x-2)}{(x+3)^2}$ $= \frac{5}{(x+3)^2}$	1 mark for correct substitution in quotient or product rule 1 mark if answer follows
A ii	$u = x^3 \quad v = (2x-9)^5$ $u' = 3x^2 \quad v' = 10(2x-9)^4$ $\frac{d}{dx} x^3 (2x-9)^5 = 3x^2 (2x-9)^5 + 10x^3 (2x-9)^4$ $= x^2 (2x-9)^4 [3(2x-9) + 10x]$ $= x^2 (2x-9)^4 (6x - 27 + 10x)$ $= x^2 (2x-9)^4 (16x - 27)$	1 mark for correct differentiation of u and v 1 mark for correct substitution into product rule 1 mark for fully factored result

B i	$\alpha + \beta = -\frac{12}{1}$ $= -12$	1 mark for correct answer
B ii	$\alpha\beta = \frac{3}{1}$ $= 3$	1 mark for correct answer
B iii	$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta}$ $= \frac{-12}{3}$ $= -4$	1 mark for $\frac{\alpha + \beta}{\alpha\beta}$ 1 mark for correct answer
B iv	From ii $\beta = \frac{3}{\alpha}$ $\therefore \alpha + \frac{3}{\alpha} = \alpha + \beta$ $= -12$	1 mark for $\beta = \frac{3}{\alpha}$ 1 mark for correct answer
C	$a(x-2)^2 + b(x+4) + 1 + c$ $= a(x^2 - 4x + 4) + b(x+4) + 1 + c$ $= ax^2 - 4ax + 4a + bx + 4b + 1 + c$ $= ax^2 + (b - 4a)x + (4a + 4b + 1 + c)$ $\therefore a = 3$ $\therefore b - 4a = -14$ $b - 12 = -14$ $b = -2$ $\therefore 4a + 4b + 1 + c = -2$ $12 - 8 + 1 + c = -2$ $5 + c = -2$ $c = -7$ $\therefore a = 3, b = -2, c = -7$	1 mark for correct expansion and factorising 1 mark for correct value of a and if b follows 1 mark if value of c follows

SECTION 3		
Q	Solution	Suggested marking criteria
A i	$f(x) = (4x + b)^{\frac{1}{2}}$ $f'(x) = \frac{1}{2}(4x + b)^{-\frac{1}{2}} \times 4$ $= 2(4x + b)^{-\frac{1}{2}}$ $= \frac{2}{(4x + b)^{\frac{1}{2}}}$ $= \frac{2}{\sqrt{4x + b}}$	1 mark for correctly showing use of the chain rule 1 mark for fully showing the result

A ii	$\frac{2}{\sqrt{4x+b}} = \frac{2}{5} \text{ when } x = 11$ $\frac{2}{\sqrt{44+b}} = \frac{2}{5}$ $\therefore \sqrt{44+b} = 5$ $44+b = 25$ $b = -19$	1 mark for $\frac{2}{\sqrt{44+b}} = \frac{2}{5}$ 1 mark if solution follows
B i	$y^2 - 4y = 6x + 8$ $(y-2)^2 = 6x + 12$ $(y-2)^2 = 6(x+2)$ Vertex is at $(-2, 2)$	1 mark for $y(y-4) = 6\left(x + \frac{4}{3}\right)$ 1 mark if vertex follows
B ii	$4a = 6$ $a = \frac{3}{2}$ Focus is at $\left(-\frac{1}{2}, 2\right)$	1 mark for focal length following from b i 1 mark if focus follows
B iii	$3x + 4y - 3 = 0$ $3x = -4y + 3$ $6x = -8y + 6$ Subs into parabola $y^2 - 4y - (-8y + 6) - 8 = 0$ $y^2 - 4y + 8y - 6 - 8 = 0$ $y^2 + 4y - 14 = 0$ $\Delta = 4^2 - 4 \times 1 \times -14$ $= 16 + 56$ $= 72$ $\Delta > 0 \therefore$ two points of intersection between the line and parabola	1 mark for $y^2 - 4y - (-8y + 6) - 8 = 0$ or equivalent 1 mark for correctly using the discriminant to show the result